

MODULE 1 – VARIABLE FREQUENCY DRIVE OPERATIONS

Q1.

Explain the criteria for selecting an appropriate inverter and its required peripheral devices based on specific motor types. Illustrate the standard power and signal connections of a VFD using a detailed block diagram.

Q2.

Describe the working principles of Sensorless Vector Control (SVC) and Permanent Magnet (PM) motor control. Discuss their comparative advantages and industrial applications for precise speed and torque regulation.

Q3.

Discuss the significance of programmable I/O assignment and pulse input functions in VFD operations. Explain how the PTC interface is integrated into the drive to provide continuous thermal protection for the motor.

Q4.

Explain the configuration of core VFD parameters required to transition between speed, torque, and position control modes. Apply this concept to describe the drive setup for an industrial material-handling or positioning system.

Q5.

Describe the calibration function and the process of establishing baseline operating parameters within a VFD. Discuss how accurate parameterization ensures optimal motor performance and prevents drive tripping under varying load conditions.

Q6.

Discuss common VFD operational faults such as overcurrent, overvoltage, and motor overload. Explain the systematic keypad-based troubleshooting procedure used to diagnose and clear these faults through the drive interface.

MODULE 2 – INTERFACING

Q1.

Explain the importance of grounding, shielding, and isolation when wiring analog I/O between a PLC and a VFD. Discuss how electromagnetic interference (EMI) is mitigated to prevent unstable motor control in industrial environments.

Q2.

Describe the master-slave architecture in a multi-drive RS-485 network. Explain the configuration of node addresses, baud rates, and parity settings required to establish reliable communication between a single PLC and multiple cascaded VFDs.

Q3.

Discuss the integration of an encoder for closed-loop vector control. Apply this concept to explain how high-speed pulse-train feedback is wired and processed by a PLC to achieve precise speed and position synchronization.

Q4.

Explain the process of analog signal scaling and mapping between a sensor, a PLC, and a VFD. Apply this concept to demonstrate how a 4–20 mA flow transmitter signal is converted into a 0–50 Hz drive output for a cooling pump.

Q5.

Describe the Safe Torque Off (STO) function integrated into modern VFD hardware interfaces. Discuss its industrial significance in meeting machine safety standards and explain how it differs from traditional line-contactor isolation.

Q6.

Discuss the integration of VFDs into higher-level automation networks using gateways or bridge modules. Explain how cyclic and acyclic data exchange is managed in a Profibus or EtherNet/IP architecture to monitor drive health and performance.

MODULE 3 – HUMAN MACHINE INTERFACE (HMI)

Q1.

Explain the principles of ergonomic UI design and color dynamics in HMI development. Discuss how effective screen layouts and visual indicators enhance operator situational awareness in complex motor-driven processes.

Q2.

Describe the configuration of dynamic graphic animations (e.g., rotating motors and filling tanks) in an HMI workspace. Explain how state tags and integer values from a PLC are mapped to control object visibility and motion.

Q3.

Discuss the role of background logic and macros (scripting) executed locally on the HMI terminal. Apply this concept to explain how an HMI can perform local unit conversions (e.g., converting VFD frequency in Hz to motor RPM) before displaying the data.

Q4.

Explain the mechanisms of historical data storage and external database connectivity (e.g., SQL) in advanced HMI panels. Discuss how this data is utilized to generate and export shift-based production reports for an automated assembly line.

Q5.

Describe the integration of HMI systems with higher-level SCADA systems or IIoT cloud platforms through web server functionalities. Explain how remote access features enable off-site engineers to perform VFD diagnostics and monitoring.

Q6.

Discuss the implementation of multi-language support and role-specific dashboards in a globally deployed industrial machine. Apply this feature to a scenario in which maintenance personnel and machine operators require different levels of data visibility and control.